



Rice production in Africa: current situation and issues

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INTRODUCTION

Rice is a staple food in many countries of Africa and constitutes a major part of the diet in many others. During the past three decades the crop has seen consistent increases in demand and its growing importance is evident in the strategic food security planning policies of many countries. With the exception of a few countries that have attained self-sufficiency in rice production, rice demand exceeds production and large quantities of rice are imported to meet demand at a huge cost in hard currency. Africa consumes a total of 11.6 million tonnes of milled rice per year (FAO, 1996), of which 3.3 million tonnes (33.6 percent) is imported. As many as 21 of the 39 rice-producing countries in Africa import between 50 and 99 percent of their rice requirements. The distribution of rice importation on a regional basis appears skewed, with the North and Central Africa regions setting the lower (1.7 percent) and upper (71.7 percent) limits, respectively.

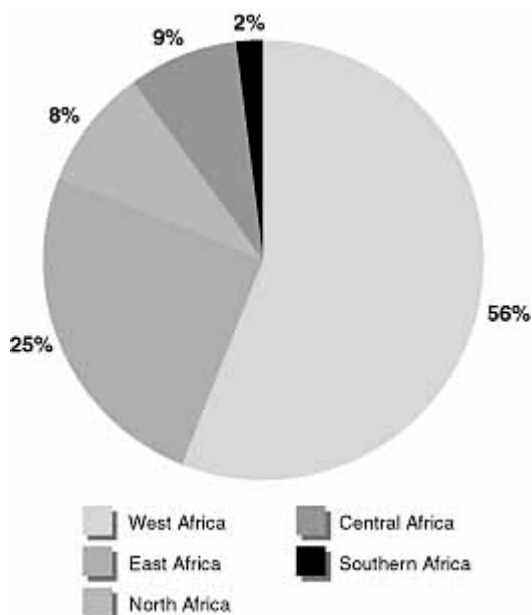
Africa's inability to reach self-sufficiency in rice is the result of several major constraints in the rice industry which require urgent redress to stem the trend of over-reliance on imports and to satisfy the increasing demand for rice in areas where the potential of local production resources is exploited at very low levels.

RICE PRODUCTION, CONSUMPTION AND SELF-SUFFICIENCY IN AFRICA

Rice production

Africa produces an average of 14.6 million tonnes of rough rice per year (1989-1996) on 7.3 million ha, equivalent to 2.6 and 4.6 percent of the world's total production and rice area, respectively. West Africa has the greatest rice area (Figure 1) in Africa (56.5 percent), i.e. about 3.7 million ha.

FIGURE 1: Rice area by region, Africa



The rice area has increased steadily from 1989-91 to 1996 (Figure 2), with a total increase of 900 000 ha in five years. Regional area trends (Figure 3) indicate that between 1989-91 and 1996 West Africa had the greatest increase in area, the four other regions showing only insignificant changes. The regional contributions to rice production in Africa, presented in Figure 4, show the following declining order: West Africa (42 percent); North Africa (32 percent); East Africa (23.8 percent); Central Africa (1.2 percent); and southern Africa (1 percent).

FIGURE 2: Rice area trends in Africa

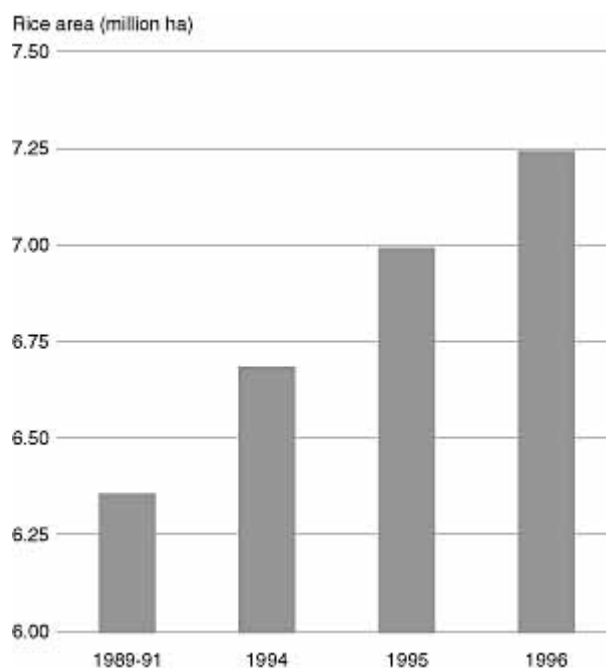


FIGURE 3: Regional trends in rice area, Africa

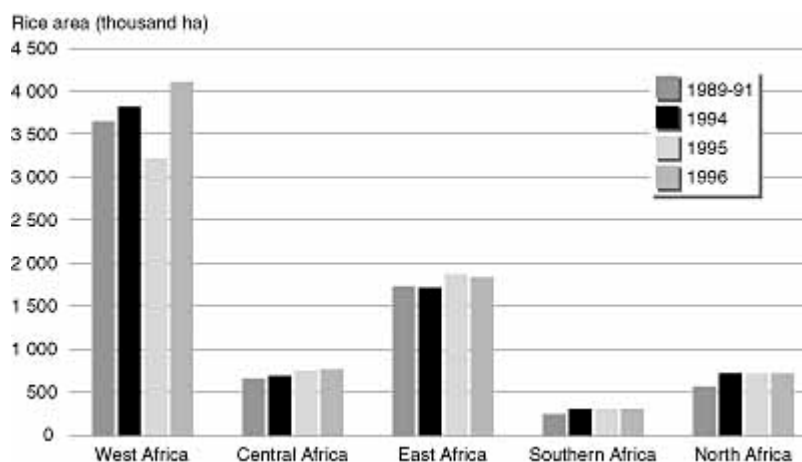
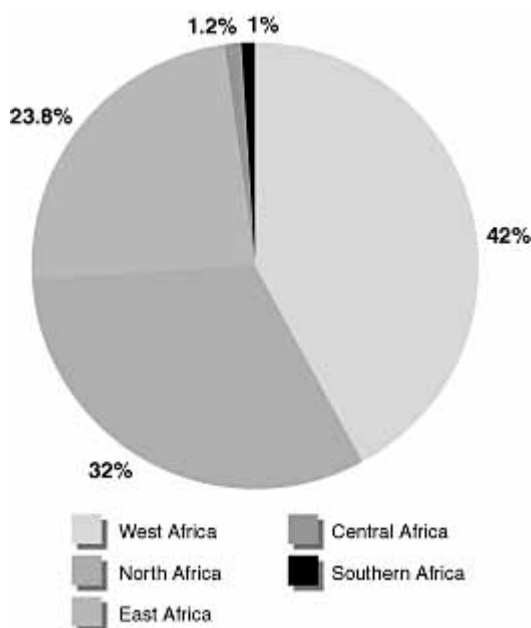
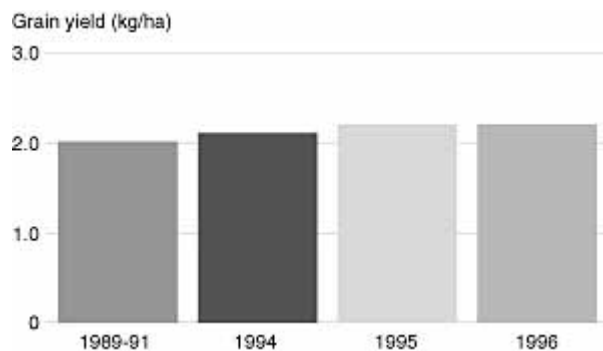


FIGURE 4: Rice production by region, Africa



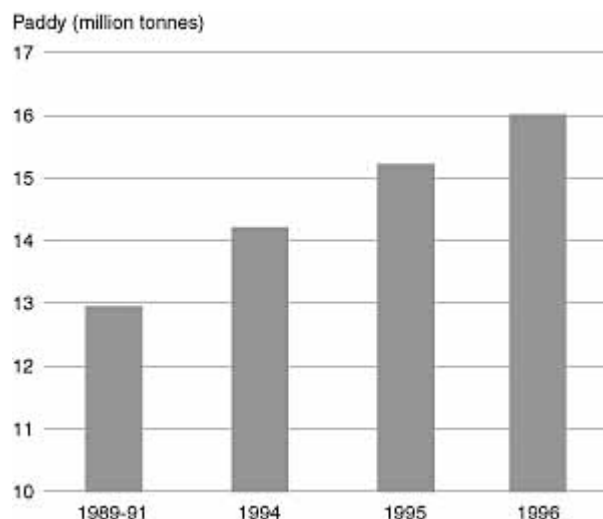
The average grain yield in Africa (2.1 kg/ha) is 49 percent below the world average (3.4 kg/ha). This low grain yield is caused by several factors, including the low levels of production technologies and the dominance of the upland ecology (55 percent) in Africa. Irrigated ecology represents only 11 percent of the rice area in Africa, compared with 53 percent worldwide (Kaung, John and Alam, 1985). The average grain yield in Africa shows very little improvement over time (Figure 5).

FIGURE 5: Grain yield trends in Africa

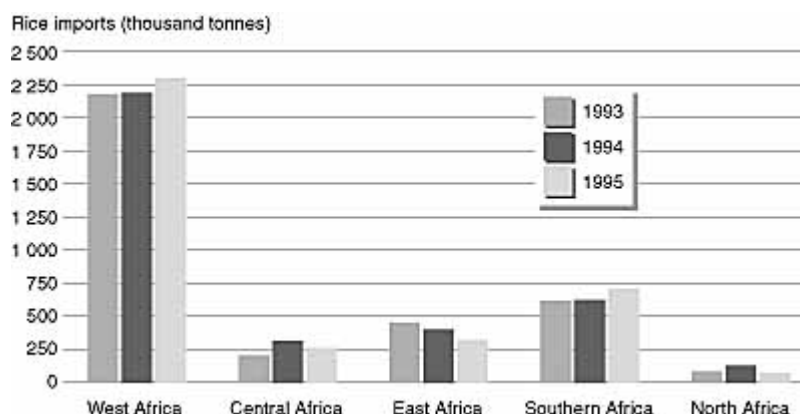


Rice consumption

Africa consumes 11.5 million tonnes of rice per year (FAO, 1996), 33.6 percent of which is imported. Between 1989-91 and 1995, the rice import trends showed a small decline of 55 000 tonnes, while paddy production increased by 2.3 million tonnes over the same period (Figure 6).

FIGURE 6: Paddy production trends in Africa

The regional trends in rice imports show that the small decline in imports could be attributed mainly to East Africa (Figure 7). The West Africa region's rice self-sufficiency levels between 1960 and 1996 are presented in Table 1, and the 1996 data for the four other regions are presented in Table 2.

FIGURE 7: Regional trends in rice imports, Africa**TABLE 1****Rice self-sufficiency (percent) in West Africa**

Country	1960-64 ¹	1970-74 ¹	1980-84 ¹	1989-96 ²
Benin	12.4	26.7	22.9	17.7
Burkina Faso	83.1	74.3	34.2	39.3
Chad	-	-	-	75.0
Côte d'Ivoire	70.1	71.3	42.0	62.8
Gambia	84.8	87.1	23.0	18.0
Ghana	28.8	48.3	41.2	15.1
Guinea	83.1	82.9	74.4	56.7
Guinea-Bissau	94.6	47.4	60.1	55.0
Liberia	63.5	71.1	64.1	38.5
Mali	100.0	69.7	45.2	48.9
Mauritania	4.2	5.3	10.0	54.1
Niger	63.7	52.8	26.7	49.6
Nigeria	98.7	98.8	51.4	83.8
Sierra Leone	92.6	91.4	77.4	55.0
Senegal	27.8	23.3	15.5	20.3
Togo	70.8	70.0	28.4	56.0

Average	72.2	71.5	48.3	46.6
¹ Nyanteng, 1998.				
² FAO, 1995.				

The average self-sufficiency levels for the West African countries showed a steady decline from 1960-64 to 1996 (i.e. from 72.2 to 46.6 percent), an indication of the region's inability to meet the increasing demand through local production. The average self-sufficiency of Central Africa reached 37.9 percent, of East Africa 58.3 percent, and of southern Africa 42.5 percent. Meanwhile, in North Africa, Egypt produces enough to meet local demand and exports the surplus, but its export market has declined because of increased local consumption. However, Egypt has complete self-sufficiency in rice (99.99 percent) (Table 2).

TABLE 2							
Rice self-sufficiency of African regions, 1996							
Central Africa	Self-sufficiency (%)	East Africa	Self-sufficiency (%)	Southern Africa	Self-sufficiency (%)	Northern Africa	Self-sufficiency (%)
Burundi	83.1	Kenya	41.0	Angola	94.9	Algeria	2.5
Cameroon	23.2	Madagascar	96.6	Mozambique	98.8	Egypt	99.99
Central African Rep.	83.5	Malawi	90.5	South Africa	0.93	Morocco	99.97
Congo	1.9	Somalia	2.9	Swaziland	99.1		
Dem. Rep. of Congo	37.6	Sudan	0.4	Zambia	100.0		
Gabon	2.0	United Rep. Tanzania	84.9	Zimbabwe	0		
Rwanda	34.2	Uganda	91.6				

Morocco also has complete self-sufficiency (99.97 percent), but it has a low rice demand (21 000 tonnes) which is met by local production and negligible imports. Algeria has a very low local production (1 200 tonnes) and imports nearly all of its consumption (46 000 tonnes).

RICE POTENTIAL IN AFRICA

The potential arable land in Africa is 637 million ha (Okigbo, 1982), and about 68 percent of the total area is in reserves. Africa therefore has great potential for expanding its agricultural production in general and rice in particular. Rice production is most extensive in rainfed (upland) ecosystems (which account for 60 percent of the total rice area) and, in these areas, rice competes with several other important staple crops, such as maize, sorghum, millet, cassava, yam, coco-yam, plantain and banana, as well as such cash crops as coffee, cocoa, citrus and cola. However, as these crops increase in economic importance, the rice area will diminish because rice is the lowest-yielding crop in that ecology. The wetlands, including irrigated ecologies, will therefore become increasingly important as these ecologies are more suitable for rice production than they are for other upland crops.

The wetlands in tropical sub-Saharan Africa cover a total area of 2.4 million km² (24 million ha) and consist of four categories (Table 3): coastal wetlands (165 000 km²); inland basins (1.07 million km²); river floodplains (300 000 km²); and inland valleys (850 000 km²) (Andriesse, 1986). Osborn (1953), quoted by Guthrie (1985), pointed out that Africa, which represents one-fifth of the world's land, is about one-third desert and only 0.1 percent of its total land area is irrigated. Wetland soils occupy a larger area than irrigated land but, compared with the total, this area is still quite small. There are, however, several large areas, principally in Central Africa, where wetland soils are common and are dominated by Tropaequpts and Tropoquents of limited suitability for conversion to productive use.

TABLE 3
Wetlands in tropical sub-Saharan Africa (2.4 million km²)

Category	Area ('000 km ²)	% of total wetlands	% of total area
Coastal wetlands	165	7	1.5
Inland basins	1075	45	9.0
River floodplains	300	12	2.5
Inland valleys	850	36	7.0
Source: Andriesse, 1986.			

Table 3 shows that the inland basins and inland valleys, which constitute 45 and 36 percent of the wetlands, respectively, have high potential for the development of rice in Africa.

Many rivers in Africa (e.g. the Gambia, Niger, Benue, Zaire, Zambia, Limpopo, Tana, White and Blue Nile and Chari rivers) have well-developed large floodplains in their central and lower stretches, and these are suitable for rice production (Andriesse, 1986).

RICE ECOSYSTEMS

Rice is produced in Africa in the following five main ecosystems:

- dryland (rainfed upland);
- hydromorphic (rainfed lowland);
- mangrove swamp;
- inland swamp;
- irrigated ecology.

The various ecosystems face many constraints, some of which are specific to particular ecosystems, while others are general and cut across ecosystems and regions.

The specific constraints will be discussed under each ecosystem, while the non-specific ones will be described generally.

Dryland rice ecosystem (strictly rainfed)

Description. This is the most extensive rice ecosystem in Africa, so it has a great influence on the total rice output. It occurs in the uppermost part of the topo-sequence and is more important in West Africa than in other African regions. In West Africa it covers 57 percent (i.e. 1.8 million ha) of the total rice area and accounts for 44 percent of regional production (Table 4). The major producers in West Africa include the following countries: Sierra Leone, Côte d'Ivoire, Liberia, Guinea-Bissau and Nigeria.

TABLE 4

Rice area, yield and production by ecosystem in West Africa (1980-84 and 2000)

Environment	Area		Yield	Production
	('000 ha)	(%)	(tonnes/ha)	('000 tonnes)
1980-84				
Upland/hydromorphic	1 490	57	1.0	1 490
Hydromorphic/inland swamp	530	20	1.4	742
Irrigated humid zone	119	5	2.8	333
Sahel	112	4	2.8	313
Mangrove swamp	193	7	1.8	347
Deepwater/floating	190	7	0.9	171
Total	2 631	100	-	3 396
2000 ¹				
Upland/hydromorphic	2 160	59	1.0	2 160
Hydromorphic/inland swamp	760	21	1.4	1 064

Irrigated humid zone	185	5	2.8	518
Sahel	155	4	2.8	434
Mangrove swamp	193	6	1.8	347
Deepwater/floating	190	5	0.9	171
Total	3 643	100	-	4 694
¹ Projections. Source: WARDA, 1993.				

Constraints. The only source of water is rain, so the crop is highly vulnerable to drought as a result of erratic and poor rains. This ecology occurs mainly in regions where the soils are highly weathered and the dominant clay is usually kaolinite with low water-retention capacity. Soils in this ecosystem are usually poor in nitrogen (N), phosphorus (P) and sulphur (S). Iron (Fe) deficiency may also occur. Phosphorus deficiency results mainly from fixation with Fe^{2+} , manganese (Mn^{2+}) and aluminium (Al^{3+}). The rainfed upland ecology is characteristically an extensive rice culture based on the traditional shifting cultivation system. The drawbacks of the system are many:

- It leads to rapid deforestation and destruction of watersheds.
- It is essentially a low-input ecosystem, which results in poor paddy yields (i.e. 1 tonne/ha) for the following reasons:
 - there is little or no fertilizer application in this ecosystem;
 - weed control is essentially manual and is usually delayed, resulting in serious yield reduction;
 - drought results in total crop failure;
 - diseases, such as blast, leaf scald, brown spot and sheath rot, cause considerable loss in yields if not controlled;
 - the major pests include birds, mice and the large rodents known as "grasscutters" (*Thryonomis swindarianus*), which cause serious destruction of rice.
- It is wasteful of land resources since increased production is achieved mainly through area expansion rather than through increased yield per unit area.
- It is ultimately unsustainable as a result of increased population pressures on the land which have led to reduced fallow periods and poor natural regeneration of soil fertility, on which the whole culture is pivoted.

In order to develop an intensive upland rice culture, it will be necessary to undertake the following measures:

- practise rice-based cropping systems that include grain legumes to improve soil fertility;
- incorporate crop residue into the soil to improve the soil organic matter content and fertility;
- encourage the use of fertilizers and herbicides to improve growth and weed control;
- adopt erosion control methods to prevent rapid soil degradation;
- adopt water-harvesting methods to reduce the risk of drought and enhance crop yields;
- evolve upland rice varieties that are drought-tolerant and high-yielding.

Lowland/hydromorphic

Description. This ecosystem occurs from the mid-slope to the valley bottom in the topo-sequence. The rice crop here may obtain water from three sources - direct rainfall, high water table and surface water - depending on its location in the topo-sequence. The main hydraulic characteristic of this ecosystem is the fluctuating water table, caused by cyclical swelling and receding water levels of rivers during the rains.

It is estimated that a total of 130 million ha of inland valleys are available for cultivation in Africa, 19 million ha of which (i.e. 14.6 percent) occur in West Africa. Depending on the level of technology, soil conditions and socio-economic factors, the range of yields is wide (between 1.4 and 5 tonnes/ha) but generally higher than in the strictly upland ecosystem (Table 4).

Constraints. As a result of poor drainage, high levels of Fe^{2+} and Mn^{2+} may occur and often produce iron toxicity symptoms. Iron toxicity has been observed in many West African countries, including Benin, Burkina Faso, Côte d'Ivoire, Liberia, Nigeria, Senegal and Sierra Leone (Virmani, 1979). Yield losses of between 12 and 88 percent have been observed as a result of iron toxicity (Gunawardena, Virmani and Sumo, 1982). Non-tolerant lines could result in 100 percent yield reductions. A major physical constraint in this ecosystem are the uncontrolled floodwaters that sometimes inundate the crop or produce flash floods, which may carry away the harvest.

Mangrove swamps

Description. Mangrove swamps occur mainly along the West African coast and cover a total area of 1.2 million ha, 193 000 ha (i.e. 16 percent) of which have been developed for cultivation (Table 4).

The mangrove swamps have high salinity levels caused by seawater intrusion brought in by tidal waves from the sea, although nearly all mangrove swamps enjoy a salt-free period during the rainy season as freshwater floods wash the land. This period shortens, from over six months to under four, with increasing proximity to the sea, but is generally long enough to allow a crop of rice to grow.

Constraints. About 84 percent of the potential area is uncultivated, but its development is likely to be very slow because of the following discouraging factors:

- the high cost of development;
- inadequate tools for development ;
- long distances between the swamps and villages making access difficult;
- shortage of labour and its attendant high cost;
- the control of mangrove clearing for ecological reasons.

The chances of mangrove swamps producing a major impact on rice production in West Africa in particular and Africa in general are, therefore, remote.

Major constraints in the mangrove swamps include:

- lodging and silting, caused by tidal movements;
- adverse soil conditions, especially acidity and salinity, as well as iron, aluminium and manganese toxicity;
- weeds, especially in the swamps further from the sea;
- pests, among which crabs are by far the most damaging, although stem borers and rice bugs are also important;
- diseases, including blast and brown spots, which can be severe.

The productivity of this ecosystem is very low, but improved technology and increased applications of inputs can raise yields from about 1 to 2.2 tonnes/ha.

Irrigated ecosystem: humid zone

Description. The irrigated ecosystem provides the best conditions for rice cultivation because of the better control of water compared with other ecologies. However, the importance of the irrigated ecosystem in rice production in Africa is relatively small (11 percent of the rice area) except in a few countries, such as Egypt (100 percent of the rice area), Madagascar (31 percent), the Niger (100 percent) and Mauritania (100 percent) (WARDA, 1993). The yield range is the highest of all the ecosystems (from 3.5 to 7 tonnes/ha).

The vast wetlands in the West and East Africa regions are yet to be fully developed. In West Africa, the irrigated ecosystem is expected to increase from 231 000 ha (1980-84) to 340 000 ha by the year 2000 (Table 4). The projection is based largely on the activities of Nigeria and Côte d'Ivoire which are developing their wetlands and irrigated ecosystem.

Constraints. The major physical constraints include nutrient deficiencies (i.e. N, P, S and zinc [Zn]), toxicities (Fe, Mn and Al) and acidity.

The biological constraints include: weed buildup (of grasses, broadleaves, sedges and wild rice); diseases (rice yellow mottle virus, blast, glume discoloration, sheath rot and bacterial leaf blight); and insects (gall midge, white stem borer and grain bugs). In Africa, 15 species of insects are considered major pests of rice and they are classified into four groups: stem borers; leaf feeders; grain suckers; and stem and leaf suckers. Most damage to rice is caused by stem borers. Additional constraints are: pests (birds, rodents and crabs); germplasm (poor input responsiveness of local varieties and slow release of improved varieties); and health hazards (buildup of vectors of malaria and schistosomiasis parasites). The average yield loss that can be attributed to insects, disease and weeds is about 33.7 percent of the potential yield (Barr, Koecher and Smith, 1975).

GENERAL CONSTRAINTS ACROSS ECOSYSTEMS

The general constraints affecting rice production in Africa include the following:

- unreliability of rainfall in the transitional zones of West, Central and East Africa;
- serious destruction of rice fields by cyclones in Madagascar (Rakotonjanahary, 1989);
- yield losses caused by low solar radiation owing to clouds during August in West Africa (Posner, 1978);
- extremes of temperature (< 20°C and > 35°C) which result in yield losses in Madagascar, Central, East and southern Africa;
- absence of suitable varieties to withstand these stresses (Yoshida, 1978; Masajo *et al.*, 1985; Kadena and Beachell, 1974; and Alluri, Mahsatra and Lawson, 1979);
- unfavourable government policies that affect agriculture in general and rice in particular with regard to the following:
 - resource allocation;
 - cropping priorities;
 - provision of adequate credits;
 - low interest rates;
 - creation of incentives and input subsidies;
 - pricing and marketing policies;
 - provision of modern milling facilities to improve rice milling quality and out-turn.

RICE DEVELOPMENT POLICIES, CONSTRAINTS AND PRODUCTION STRATEGIES IN AFRICA

The national policies on agricultural food production across many African countries explicitly or implicitly address the need to increase food production to meet the food security objectives of each country and give priority to achieving self-sufficiency in staple food production (e.g. cereals, and roots and tubers) in each country. The level of rice developments in the regions is closely related to:

- the status of the crop as a staple (nearly all the major rice-producing countries, i.e. Egypt, Madagascar, Sierra Leone, Senegal, Côte d'Ivoire, Liberia, Guinea and Nigeria, consume rice as a staple);
- the status of the crop as a major cash crop;
- the suitability of the prevailing ecosystem for large-scale rice production.

The following are positive factors for regional rice development:

- It is generally accepted that rice is a major food crop whose importance as a staple is rapidly growing in Africa.
- Africa imports nearly 40 percent of its rice requirements at a cost of US\$1.1 billion per year.
- Africa has vast areas of land that are suitable for rice cultivation but are currently underutilized.
- Africa has large bodies of water that could be developed for rice irrigation.

- The climate of Africa, except for a few cold and some extremely hot environments, is generally suitable for the cultivation of rice.
- Africa has a vast potential to cultivate rice for local consumption and export if large investments are made to develop the industry, reduce imports and save scarce foreign exchange.

Rice development strategies

Government agricultural policies should be directed towards increased growth of the agricultural sector and increased agricultural contributions to the gross domestic product (GDP). This will require the allocation of more government resources to agriculture and the implementation of policies aimed at solving the problems associated with technical, socio-economic, macro- and micro-economic constraints, with the following objectives:

- low-interest credit facilities for farmers;
- easy access to inputs at affordable prices, by reducing import duties on agricultural inputs and eliminating intermediaries;
- improved market infrastructures (i.e. markets, transport, roads and storage);
- improved roads and transport facilities within the farming areas to ease movement of labour, which is usually in short supply during the peak farming periods, and to prevent women, who are key personnel in many farming communities, from having to walk long distances to farms;
- infrastructures (e.g. modern mills) for better post-harvest handling of paddy, which will increase milling out-turn and the quality of the product and lead to better prices and incomes for producers;
- adequate funding for research and extension institutions, to support farmers as they increase their output, through the following interceptions:
 - development of improved high-yielding varieties for farmers;
 - optimization of the use of agro-inputs (e.g. fertilizer and herbicides);
 - development of simple farm tools to help farmers reduce the tedium of the numerous labour-intensive activities involved in the production of rice.

Short-term production strategies. In order to achieve the greatest possible impact on production in the short term it is necessary to adopt the following strategies:

- rapid expansion of the area covered by the most favourable ecology or ecologies (e.g. inland valleys, rainfed/lowland/hydromorphic, etc.);
- implementation of efficient, up-to-date production technologies (i.e. improved seeds, optimum fertilizer rates, weed control methods, etc.);
- improvement of post-harvest handling (i.e. drying, milling and packaging for market).

Long-term production strategies. In order to ensure a long-term, strong and sustainable production base, it is necessary to strengthen the research and extension facilities and overcome the major biophysical constraints in ecosystems in order to achieve the following results:

- successful breeding of varieties that fit the major production ecologies and suit consumers' tastes;
- higher yields per unit area in all of the ecosystems;
- intensification of the wetland ecosystem and consequent reduction of the area to stem the system of shifting cultivation which is becoming increasingly unsustainable;
- development of small-scale irrigation schemes, which are simple and inexpensive to construct, especially in the vast inland valleys that are found all over Africa which, generally, have higher fertility levels and better than average water availability compared with the uplands;
- improved price competitiveness for local rice against high-quality imports from the United States and Thailand by providing modern mills equipped with paddy separators and rubber rollers for higher milling out-turns (65 to 70 percent) than currently obtained (50 to 65 percent) in many countries.

Rice development support services

Coordinating rice research programmes in Africa. Rice research in Africa is organized on a subregional basis and enjoys assistance from the International Agricultural Research Centres (IARCs) such as the West African Rice Development Association (WARDA), the International Institute of Tropical Agriculture (IITA), IRRI and the International Centre of Insect Physiology and Ecology (ICIPE), which collaborate with the National Agricultural Research Systems (NARS) of the various countries.

Varietal development programmes in Africa have been greatly supported by IITA and IRRI (e.g. IRTP-Africa and INGER) collaborative studies. A WARDA-coordinated collaborative study with INGER and NARS is presently seeking control of rice yellow mottle virus at Gagnoa (Côte d'Ivoire) in sub-Saharan Africa (Sy, Alluri and Akator, 1994).

Detailed characterization of the various ecosystems and effective coordination of research within the region will avoid expensive repetitions and promote rapid development of the rice industry.

Future research needs. In order to enhance rice production, while preserving the integrity of the environment, there should be intensification of research in the following areas:

- control of rice yellow mottle virus;
- control of the African rice gall midge;
- development of an effective rice-based cropping system to improve soil fertility and pest and disease control, together with intensification of the wetland ecology to stem rapid deforestation;
- breeding of suitable varieties for cold tolerance and heat tolerance;
- increased use of organic manures to reduce overdependence on inorganic fertilizers;
- post-harvest processing to improve milling out-turn and quality of the product.

CONCLUSIONS

Among the 39 African countries that produce and consume rice, only ten have attained any appreciable levels of rice self-sufficiency (75 to 99.9 percent), while the remaining 29 are heavy importers with self-sufficiency levels ranging between 0 and 62.8 percent. Based on WARDA statistics (1973-1992), the projected rice demand by the year 2000 will be 13.08 million tonnes, representing an increase of 1.94 million tonnes since 1996. The generally low production technology practised in Africa results in low yields, which are exacerbated by the several constraints discussed. Unless Africa invests more in research to find appropriate technologies that will remove several of the major constraints to rice production in the various ecologies, the region will have to increase imports to satisfy the growing demand for rice that is expected in the twenty-first century.

Since the upland/hydromorphic ecology is the most extensive in Africa and the lowest-yielding, a 30 to 40 percent increase in yield in these areas will have a great impact on production, so their improvement should be a short-term priority. In the long term, Africa should consider the lowland (rainfed) ecology and the vast potential of the inland valleys for intense development. Because of the high cost of development and the generally poor record of irrigation management in Africa, it is prudent to avoid large-scale conventional schemes and concentrate on small irrigation units. These are simpler to manage and could be built around small village communities, thus avoiding labour shortage problems. While seeking to expand rice production, grain cooking qualities should also be improved. Varietal improvement programmes should meet the tastes and demands of the consumers who are the ultimate judges of the success of the programmes.

In the free market system, where local products compete with high-quality imports, farmers' incomes will be determined by the quality of their produce. In this regard, efforts should be made to improve harvesting methods and post-harvest processing by discarding obsolete mills and introducing modern rubber-roller mills for high out-turn and a high percentage of head rice.

Last, but not least, the onus is on governments to provide the necessary enabling environments through national agricultural policies and by providing more resources for the development and expansion of the industry in order to achieve a high level of rice self-sufficiency.

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Production de riz en Afrique: situation et problèmes actuels

La production et la consommation de riz en Afrique ne cessent d'augmenter. L'Afrique produit en moyenne 14,6 millions de tonnes de paddy (1989-1996) sur une superficie d'environ 6,8 millions d'hectares, ce qui représente 2,9 pour cent et 4,8 pour cent de la production et de la superficie mondiales, respectivement. La superficie consacrée à la riziculture représente 1,6 pour cent environ du potentiel de terres arables en Afrique. L'Afrique importe en moyenne 3,9 millions de tonnes de riz par an, représentant 30,7 pour cent de la quantité totale de riz commercialisée en Afrique, pour une valeur d'environ 1,1 milliard de dollars EU. Les ratios d'autosuffisance en riz sont faibles en Afrique de l'Ouest et en Afrique centrale (42,5 et 28,3 pour cent, respectivement), mais élevés en Afrique du Nord, en Afrique de l'Est et en Afrique australe (98,3 pour cent, 83,3 pour cent et 65,6 pour cent, respectivement).

Cinq grandes écologies du riz sont identifiées en Afrique, présentant un large éventail de contraintes associées à des facteurs physiques, chimiques et biologiques, tels que température froide ou chaude, vents secs, teneurs élevées en sulfate acide, salinité, toxicité Fe^{2+} et carence en zinc; lenteur de la mise au point de variétés de riz amélioré, présence d'herbes adventices nuisibles (comme *Echinochloa* sp, *Ischaemum rugosum*, etc.), principaux ennemis des cultures (cécidomyie du riz, oiseaux, rongeurs et crabes) et maladies telles que la pyriculariose, la pourriture et la mosaïque jaune du riz. Les politiques des gouvernements africains concernant le taux de croissance attendu du secteur agricole et son impact sur le PNB, ainsi que d'autres mesures connexes comme les priorités en matière de production (cultures d'exportation, cultures vivrières et cultures de substitution), l'allocation des ressources, les politiques de fixation de prix, les subventions, les facilités de crédit, les incitations à la production, la nature des structures de commercialisation, le niveau de développement des infrastructures (routes, installations de stockage et d'usinage modernes) et les taux d'intérêt, ont eu un impact profond sur la production de riz dans nombre de pays d'Afrique.

La nécessité de procéder à une caractérisation détaillée de tous les écosystèmes, afin de faciliter les transferts de technologies entre régions, est à l'étude ainsi que l'importance d'une coopération étroite entre les SNRA et les ICARS en vue de renforcer les capacités nécessaires pour accroître l'efficacité de la production et réduire l'écart important entre les rendements potentiels et effectifs, l'objectif étant d'accélérer la progression vers l'autosuffisance en matière de riz de l'Afrique.

Producción de arroz en África: situación y problemas actuales

La producción y consumo de arroz en África registra un aumento constante. África produce por término medio 14,6 millones de toneladas de arroz cáscara (1989-1996) en una superficie de unos 6,8 millones de ha, que equivalen al 2,9 por ciento y 4,8 por ciento, respectivamente, de la producción total y de la superficie productiva mundiales. La superficie dedicada al cultivo del arroz representa alrededor del 1,6 por ciento de la superficie arable potencial de toda África. El continente importa una media de 3,9 millones de toneladas de arroz anuales, lo que representa el 30,7 por ciento del arroz total que se comercializa en el continente africano, con un valor de aproximadamente 1 100 millones de dólares EE.UU. Los coeficientes de autoabastecimiento de arroz son bajos en el África occidental y el África central (42,5 por ciento y 28,3 por ciento, respectivamente) pero elevados en África del Norte, África oriental y África austral (98,3 por ciento, 83,3 por ciento y 65,6 por ciento, respectivamente).

En África se identifican cinco ecologías arroceras importantes, con toda una gama de limitaciones asociadas con factores físicos, químicos y biológicos, a saber, el frío, temperaturas elevadas, vientos secos y polvorientos; condiciones de sulfato ácido, salinidad, toxicidad por Fe^{2+} y deficiencia de Zn; el lento desarrollo de variedades mejoradas de arroz, presencia de malas hierbas dañinas (p. ej., *Echinochloa* sp, *Ischaemum rugosum*, etc.), grandes plagas (p. ej., la mosquilla gállica africana del arroz, las aves, los roedores y los cangrejos), y enfermedades, p.

ej., el añublo, la podredumbre de la vaina y la virosis del abigarrado amarillo. Las políticas de los gobiernos africanos en relación con la tasa prevista de crecimiento del sector agrícola y sus efectos en el PIB y otras medidas conexas, como las prioridades de producción (p. ej., cultivos de exportación, cultivos alimentarios y cultivos sustitutivos de las importaciones), la asignación de recursos, las políticas de precios, las subvenciones, las facilidades crediticias, los incentivos a la producción, la naturaleza de las estructuras comerciales, el nivel de desarrollo infraestructural (p. ej., caminos, instalaciones modernas de almacenamiento y de elaboración) y los tipos de interés, son factores todos ellos que han tenido hondos efectos en la producción arrocerá de muchos países africanos.

Se contempla la necesidad de realizar una caracterización minuciosa de todos los ecosistemas para favorecer las transferencias tecnológicas entre las regiones, así como la importancia de una colaboración estrecha entre los SNIA y los CIIA para la creación de la capacidad necesaria a fin de incrementar la eficacia productiva y reducir la amplia diferencia existente entre el rendimiento potencial y el real, con objeto de acelerar el ritmo de consecución de la autosuficiencia de arroz en África.

